

Monte Carlo Simulation of Portfolio Risk

An Analysis of Diversification, VaR and Expected Shortfall Using R

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BSc Actuarial Science and Risk Management

Independent Project | September 2026

1. Introduction and Objective

This project uses Monte Carlo simulation in R to investigate the risk of a £10,000 portfolio made up of Apple, Microsoft, JPMorgan Chase and Johnson & Johnson. Historical share-price data are used to estimate the returns, volatility and relationships between the four stocks.

The main aim is to simulate possible portfolio outcomes over a one-year period and use these simulations to measure risk. Value at Risk (VaR) and Expected Shortfall (ES) are calculated to examine potential losses in more extreme outcomes.

The project also looks at the benefits of diversification and how portfolio risk changes when market conditions become more stressful. Finally, some of the assumptions and limitations of the model are considered, including whether normally distributed returns provide a good representation of real stock returns.

2. Portfolio and Historical Data

Company	Ticker	Weight
Apple	AAPL	30%
Microsoft	MSFT	25%
JPMorgan Chase	JPM	25%
Johnson & Johnson	JNJ	20%
Total		100%

The four companies were selected from different areas of the market to provide some diversification within the portfolio. Apple and Microsoft provide technology exposure, JPMorgan Chase represents financial services, while Johnson & Johnson provides healthcare exposure. As the stocks are not perfectly correlated, combining them can reduce overall portfolio volatility compared with holding the assets individually.

Historical adjusted share prices were obtained using the “quantmod” package in R for the period from September 2021 to September 2026. Daily arithmetic returns were calculated from these prices and used to estimate the mean returns, volatilities, covariances and correlations between the four stocks. These estimates form the basis of the Monte Carlo simulation.

2.1 Historical Return Statistics

Stock	Mean Annual Return	Annual Volatility
AAPL	20.35%	27.97%
MSFT	15.38%	28.12%
JPM	20.60%	24.35%
JNJ	13.14%	17.55%

Microsoft had the highest historical annual volatility at 28.12%, closely followed by Apple at 27.97%, while Johnson & Johnson had the lowest at 17.55%. The differences in returns and volatility between the stocks, together with the relationships between their returns, affect the overall risk of the portfolio.

2.2 Correlation Between Stock Returns

	AAPL	MSFT	JPM	JNJ
AAPL	1.00	0.53	0.36	0.14
MSFT	0.53	1.00	0.30	-0.01
JPM	0.36	0.30	1.00	0.18
JNJ	0.14	-0.01	0.18	1.00

The correlation matrix shows that the stock returns are not perfectly correlated, which provides diversification benefits within the portfolio. Apple and Microsoft had the highest correlation at 0.53, which is reasonable given that both companies operate within the technology sector. The other correlations were generally lower, with Microsoft and Johnson & Johnson having a correlation close to zero. Combining assets that do not move perfectly together helps reduce the overall volatility of the portfolio.

3. Methodology

3.1 Annualising Historical Data

Daily returns were calculated from the historical adjusted share prices. As the simulation considers portfolio risk over a one-year period, the daily estimates were converted to annual values using 252 trading days per year. Mean daily returns were multiplied by 252, while daily volatilities were multiplied by the square root of 252. The daily covariance matrix was also multiplied by 252.

3.2 Portfolio Risk and Return

The expected portfolio return was calculated as the weighted average of the expected returns of the four stocks. Portfolio variance was calculated using the asset weights and covariance matrix, allowing the relationships between the stock returns to be included in the risk calculation.

3.3 Monte Carlo Simulation

A multivariate Monte Carlo simulation was then used to generate 10,000 possible one-year outcomes for the four stocks. Rather than simulating each stock independently, Cholesky decomposition was applied to the covariance matrix so that the simulated returns maintained the relationships observed in the historical data.

Each simulated set of stock returns was combined using the portfolio weights to produce a simulated portfolio return. This was then applied to the initial £10,000 investment to obtain a distribution of possible portfolio values. This distribution was used to estimate the portfolio's downside risk.

4. Monte Carlo Simulation

4.1 Simulated Portfolio Outcomes

Using the estimated annual returns and covariance matrix, 10,000 possible one-year portfolio outcomes were simulated. Each simulation represents a possible set of returns for the four stocks, while preserving the historical covariance structure between them.

Starting with an initial portfolio value of £10,000, the simulated returns produced a range of possible values after one year. The average simulated final portfolio value was approximately £11,769, which was close to the value implied by the portfolio's expected annual return. Around 16% of the simulated outcomes resulted in the portfolio finishing below its initial £10,000 value.

The spread of the simulated outcomes demonstrates the uncertainty surrounding the expected return. Although the portfolio has a positive expected return, there remains a meaningful possibility of experiencing a loss over a one-year period.

Measure	Result
Initial Portfolio Value	£10,000
Simulations	10,000
Expected Annual Return	17.73%
Annual Portfolio Volatility	17.60%
Average Simulated Final Value	£11,769
Probability of Loss	≈16%

5. Risk Analysis

5.1 Value at Risk

Value at Risk (VaR) was used to measure the potential downside risk of the portfolio. At the 95% confidence level, the simulated portfolio had a one-year VaR of **£1,127.37**. This means that, under the assumptions of the model, 95% of simulated outcomes produced a loss of no more than £1,127.37, while 5% produced a greater loss.

Relative to the initial £10,000 portfolio, this represents a potential loss of approximately **11.27%** at the 95% VaR threshold.

$$VaR_{95\%} = \text{95th percentile of portfolio losses}$$

5.2 Expected Shortfall

VaR identifies the point at which the worst 5% of simulated losses begin, but it does not show how severe losses may become beyond this point. Expected Shortfall (ES) was therefore also calculated.

The 95% Expected Shortfall was £1,817.35. This represents the average loss among the worst 5% of simulated outcomes. Expected Shortfall therefore provides additional information about the severity of losses in the tail of the distribution.

Risk Measure	Result
VaR (95%)	£1,127.37
Expected Shortfall (95%)	£1,817.35

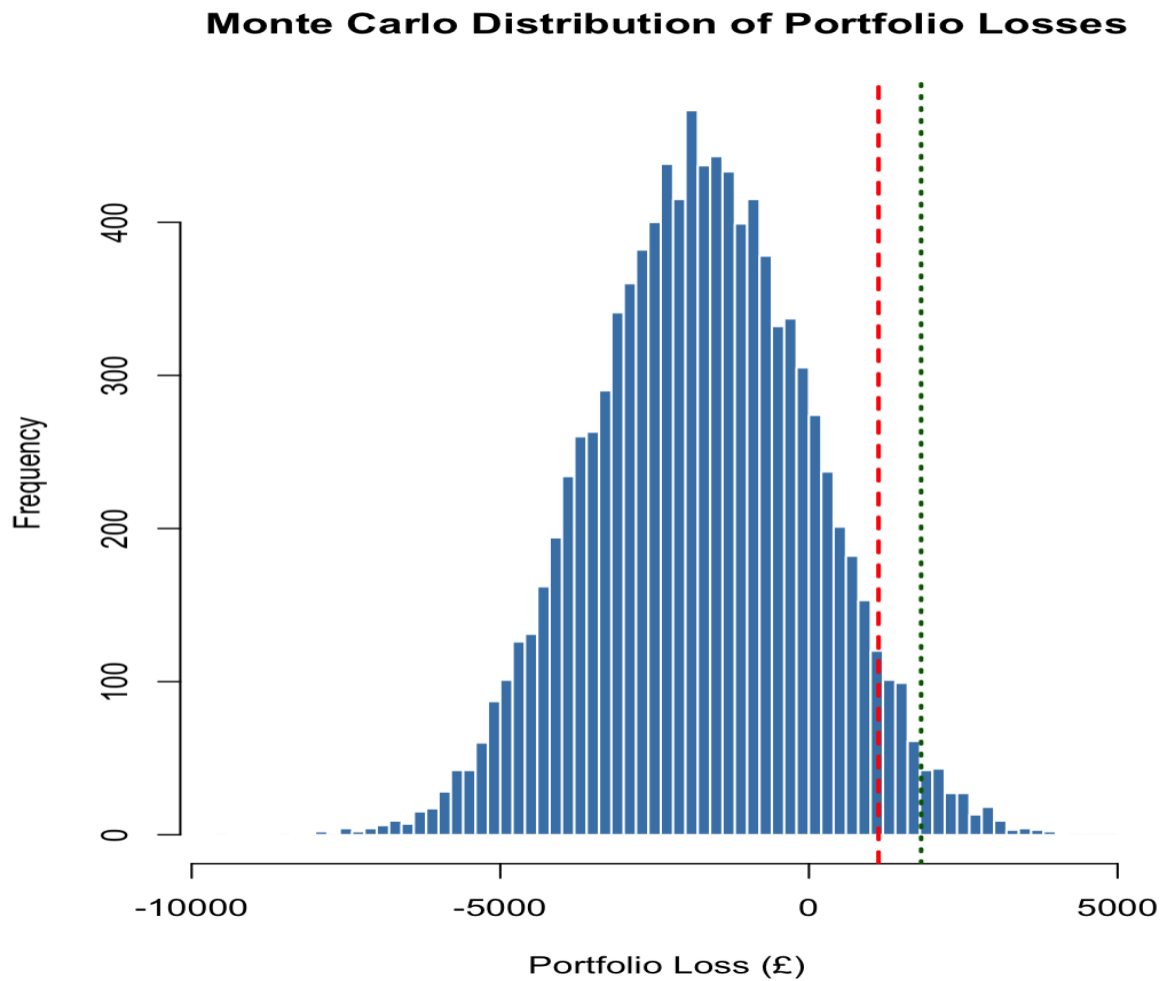


Figure 1: Monte Carlo distribution of simulated portfolio losses showing 95% VaR and Expected Shortfall.

6. Stress Testing

6.1 Volatility Test

To investigate how the portfolio would respond to more volatile market conditions, a stress scenario was created by multiplying the annual covariance matrix by 1.5. This increases the variances and covariances of the stock returns while leaving the correlation structure unchanged.

Under this scenario, annual portfolio volatility increased from 17.60% to 21.56%. The 95% VaR increased from £1,127.37 to £1,779.21, while Expected Shortfall increased from £1,817.35 to £2,624.25. This demonstrates how an increase in market volatility can substantially increase the potential size of portfolio losses.

6.2 Correlation Stress

A second stress scenario was used to investigate the effect of reduced diversification. The pairwise correlations between all four stocks were increased to 0.8, while their individual volatilities were kept unchanged.

Portfolio volatility increased from 17.60% to 23.12%, despite the individual stock volatilities remaining unchanged. The 95% VaR increased to £2,010.84, while Expected Shortfall increased to £2,954.25. This shows that higher correlations can significantly increase portfolio risk by reducing the benefits of diversification.

Scenario	Portfolio Volatility	VaR (95%)	ES (95%)
Original	17.60%	£1,127.37	£1,817.35
Volatility Stress	21.56%	£1,779.21	£2,624.25
Correlation Stress	23.12%	£2,010.84	£2,954.25

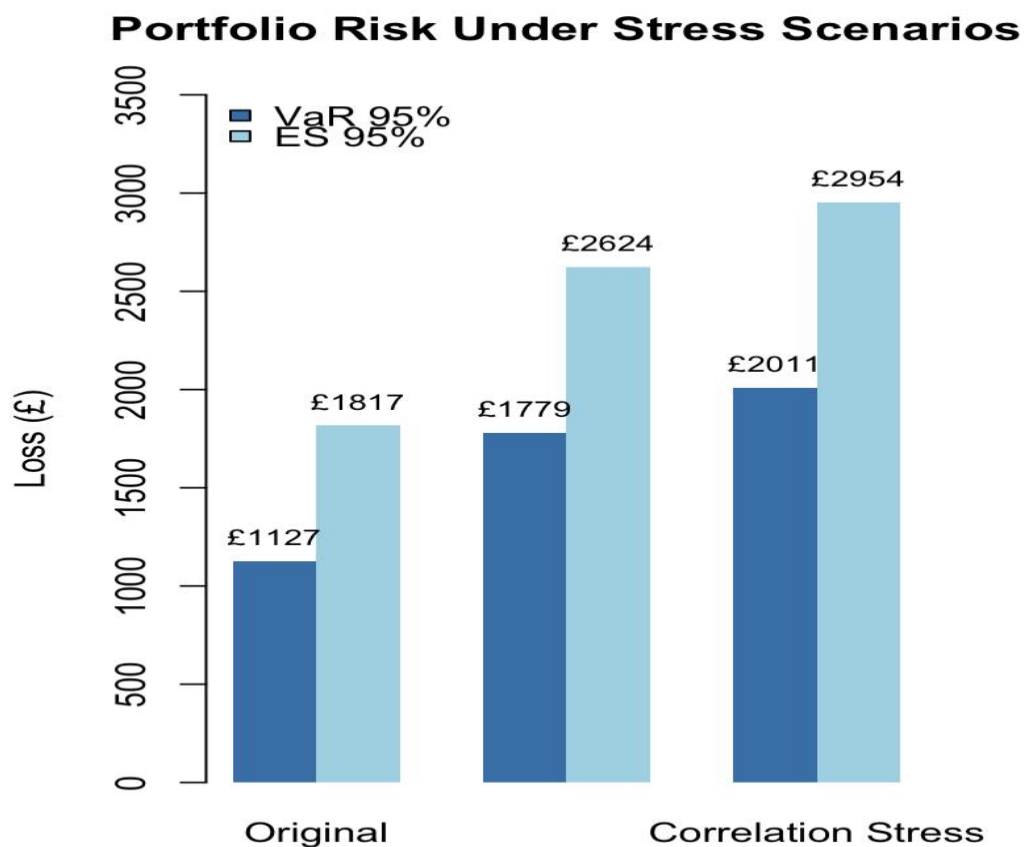


Figure 2. Comparison of 95% VaR and Expected Shortfall under the original and stressed portfolio scenarios.

7. Portfolio Path Simulation

7.1 One-Year Portfolio Paths

To show how the portfolio value could develop throughout the year, 500 Monte Carlo paths were simulated over 252 trading days. Each path began with the same initial investment of £10,000 and used daily expected returns and the historical covariance matrix to simulate the movement of the four stocks.

The simulated paths become increasingly spread out over time, reflecting the uncertainty in future portfolio values. Although the simulations generally show an upward tendency due to the positive historical mean returns, some paths finish below the initial £10,000 investment.

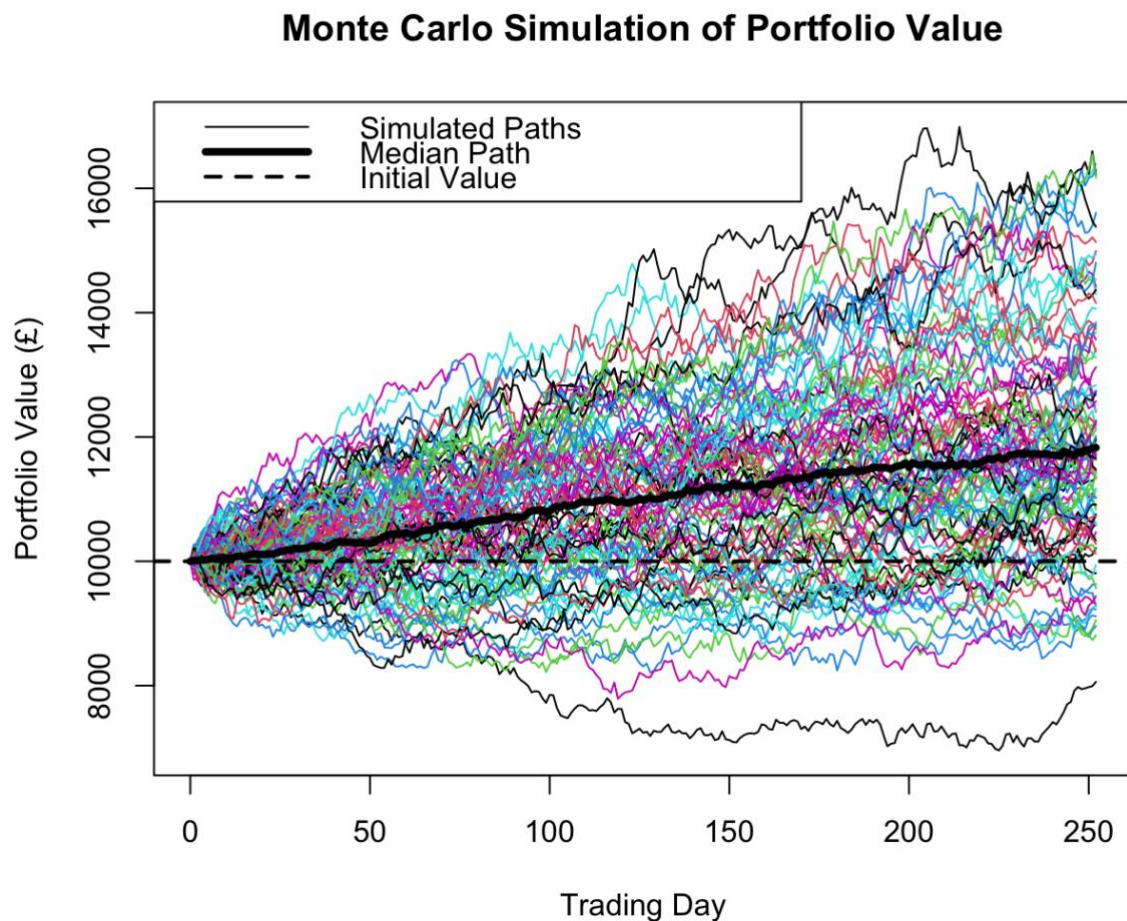


Figure 3. Monte Carlo simulation of one-year portfolio values over 252 trading days.

7.2 Final Portfolio Values

Across the 500 simulated paths, the mean final portfolio value was £12,024, while the median was £11,828. Approximately **15.2%** of the simulated paths finished below the initial £10,000 investment.

The 5th and 95th percentiles of the final portfolio values were approximately **£9,049 and £15,526**, respectively. Therefore, 90% of the simulated final values fell within this range. The mean final value was higher than the median because larger positive outcomes increased the arithmetic average.

Measure	Result
Mean Final Value	£12,024
Median Final Value	£11,828
Probability of Loss	15.2%
5th Percentile	£9,049
95th Percentile	£15,526

8. Model Assumptions and Limitations

8.1 Normal Distribution Assumption

The Monte Carlo model assumes that stock returns follow a multivariate normal distribution. This makes it possible to simulate correlated returns using the estimated mean returns, volatilities and covariance matrix. However, actual stock returns may not follow a normal distribution, particularly during periods of significant market movements.

To investigate this assumption, the historical daily returns of Apple were compared with a normal distribution using a Q-Q plot. While the observations were reasonably close to the theoretical Normal distribution around the centre, noticeable deviations occurred in both tails. This suggests that extreme returns occurred more frequently than would be expected under a Normal distribution.

Normal Q-Q Plot of AAPL Daily Returns

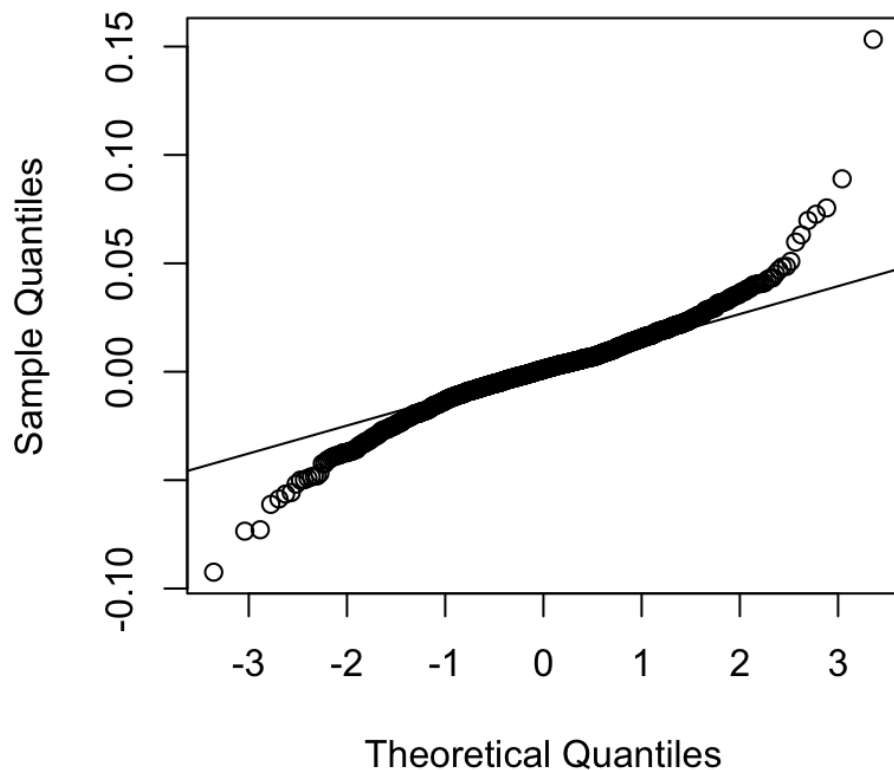


Figure 4. Normal Q-Q plot of historical AAPL daily returns.

8.2 Evidence of Heavy Tails

The Q-Q plot suggests that the historical returns have heavier tails than would be expected under a normal distribution. This is also supported by the kurtosis of Apple's daily returns, which was calculated as 9.59, compared with **3** for a normal distribution.

This indicates that extreme returns occurred more frequently in the historical data than the Normal model would suggest. As a result, the Monte Carlo simulation may underestimate the frequency and severity of extreme losses. This is particularly important when considering tail risk measures such as VaR and Expected Shortfall.

8.3 Other Model Limitations

The model relies on approximately five years of historical data to estimate expected returns, volatility and correlations. However, market conditions can change over time, meaning that these historical estimates may not accurately represent future behaviour. Correlations between stocks may increase during periods of market stress, reducing the benefits of diversification.

There is also uncertainty in the Monte Carlo simulation itself. Increasing the number of simulations reduces random simulation error and produces more stable estimates, but it does not remove uncertainty caused by the assumptions and historical parameters used in the model. Therefore, the VaR and Expected Shortfall results should be viewed as estimates under the chosen model rather than exact predictions of future losses.

9. Conclusion

This project used Monte Carlo simulation to investigate the risk of a £10,000 portfolio consisting of Apple, Microsoft, JPMorgan Chase and Johnson & Johnson. Historical market data were used to estimate expected returns, volatility and correlations, with Cholesky decomposition used to preserve the relationships between the stocks when simulating future returns.

The portfolio had an estimated annual volatility of 17.60%, with a 95% VaR of £1,127 and Expected Shortfall of £1,817. Stress testing showed that portfolio risk increased considerably when volatility or correlations were increased. Increasing correlations to 0.8 raised portfolio volatility to 23.12%, demonstrating the importance of diversification and how its benefits can be reduced when assets move more closely together.

The analysis also highlighted the limitations of the model. Historical returns showed evidence of heavier tails than the Normal distribution, meaning that a Normal-based Monte Carlo model may underestimate extreme market movements. The results are also dependent on historical estimates that may not represent future market conditions.

Overall, the project demonstrated how Monte Carlo simulation can be used alongside VaR, Expected Shortfall and stress testing to assess portfolio risk, while also showing the importance of understanding the assumptions behind the model.